

Prediction Competition

ECON 594, Fall 2026. Due Thursday 11/5/2026.

1 The problem

You get the sale records of a Japanese woodblock print auction house, <https://www.fujiarts.com/>: the artist, the title, the series, the condition the house reported, whether it is an original or a reproduction, and the date. Your assignment is to write a model that predicts **the natural log what it sold for** (you need to apply the log transformation yourself).

12,946 auctions are available in the data. Another 3,222 exist but are not included in the data. I run your code on my machine against those held-back rows.

2 The four files you get

<code>train.csv</code>	12,946 lots that sold, with <code>final_bid</code> .
<code>test.csv</code>	The 3,222 held-back lots. Every column except the price.
<code>sample_submission.csv</code>	<code>test.csv</code> with a <code>prediction</code> column already filled in, so you can see what to hand back.
<code>data_dictionary.md</code>	Information on the columns.

3 What you hand in

Build a model on `train.csv`, predict every row of `test.csv`, and put three things in a folder with your name on it:

<code>predictions.csv</code>	<code>test.csv</code> with a <code>prediction</code> column added on the end. Same rows, same order, one extra column.
your code	whatever produced those predictions. I read it.
<code>report.pdf</code>	four pages.

4 SSR

The sum of squared residuals:

$$\text{SSR} = \sum_i (\hat{y}_i - y_i)^2 \quad \text{where } y = \ln(\text{price})$$

Your goal will be to minimize the sum of squared residuals (or pseudo-residuals) for the natural log of the final price of the auction for the held out data.

The comparison is the laziest model available, predicting the average y for every lot:

$$\text{share you removed} = 1 - \frac{\text{your SSR}}{\text{SSR from predicting the average } y}$$

Your position against the rest of the class has no effect here. If everyone removes more than 0.75, everyone gets 40.

Share you removed	Points
above 0.75	40
0.66 to 0.75	34
0.56 to 0.66	28
0.30 to 0.56	20
0.00 to 0.30	10
below 0.00	0

5 The other 60 points

Validation and method, 20. Explain your validation and method.

Additional Sourcing, 15. Bring in anything: artist biographies, auction results elsewhere, art indices, exchange rates. Crawling and merging is encouraged and marked. Every added variable goes in a table giving the variable, what it measures, the source, and the date you collected it. An unsourced variable is an Honor Code matter. **One piece of data is off limits: the actual prices for the auctions.**

Report, 15. Approximately four pages. What you tried including what failed, what you kept and why, and what your model says about the market. Report appropriate output for the model you chose.

Reproducibility, 10. Your code, regenerates your `predictions.csv`. I will run it and review it.

6 Rules and dates

Work alone. Discuss ideas with anyone; share no code, features or predictions. AI is permitted, as is any package, including ones we did not use. Disclose what you used. You must be able to explain and rerun every line, and I may ask you to walk me through it. Do not try to obtain the held-back prices. No late submissions.

Data released	Wednesday 09/30/2026
Undergraduates: tell me if you are sitting Test 2 instead	Thursday 10/29/2026
Submissions close	Thursday 11/5/2026
